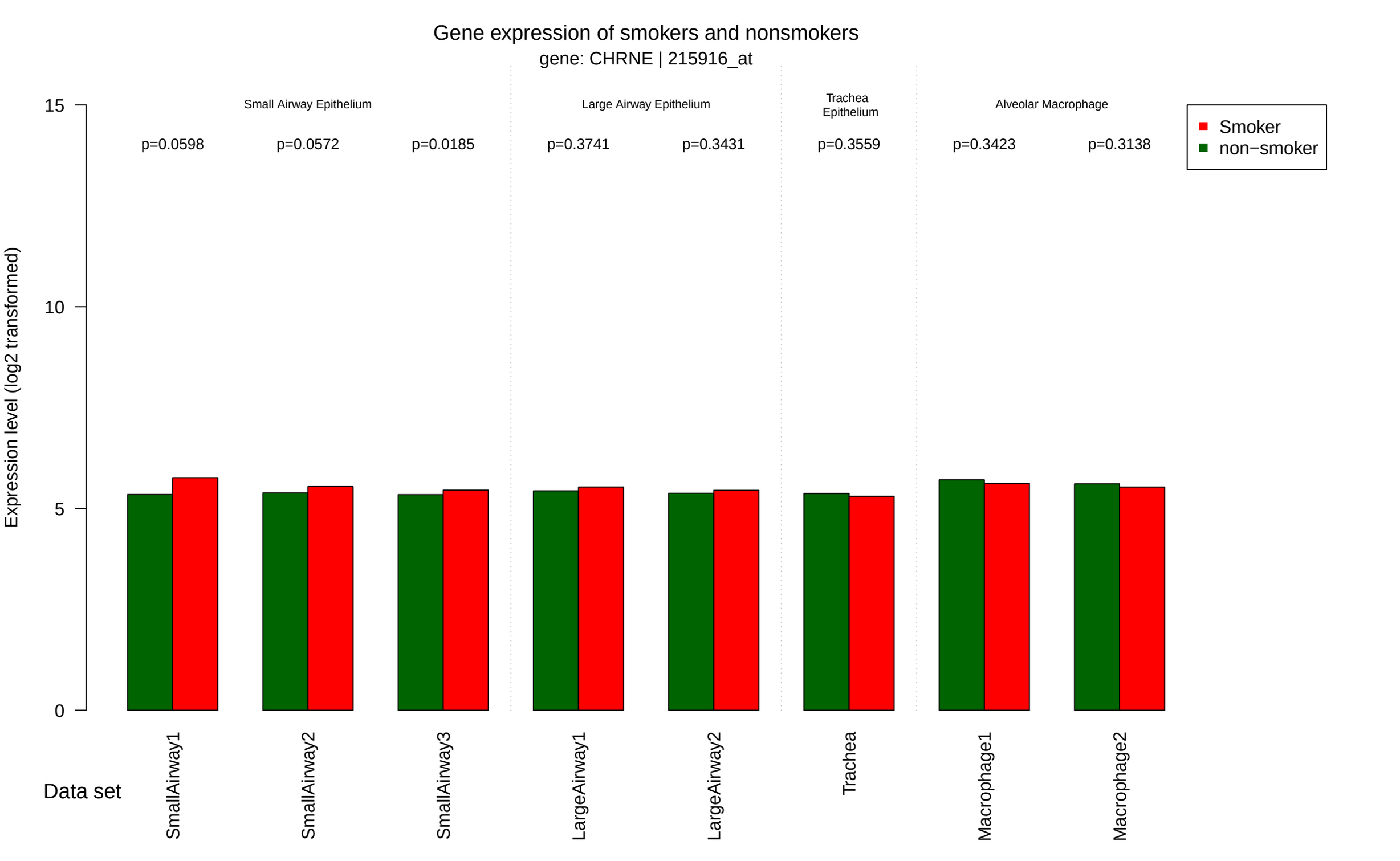
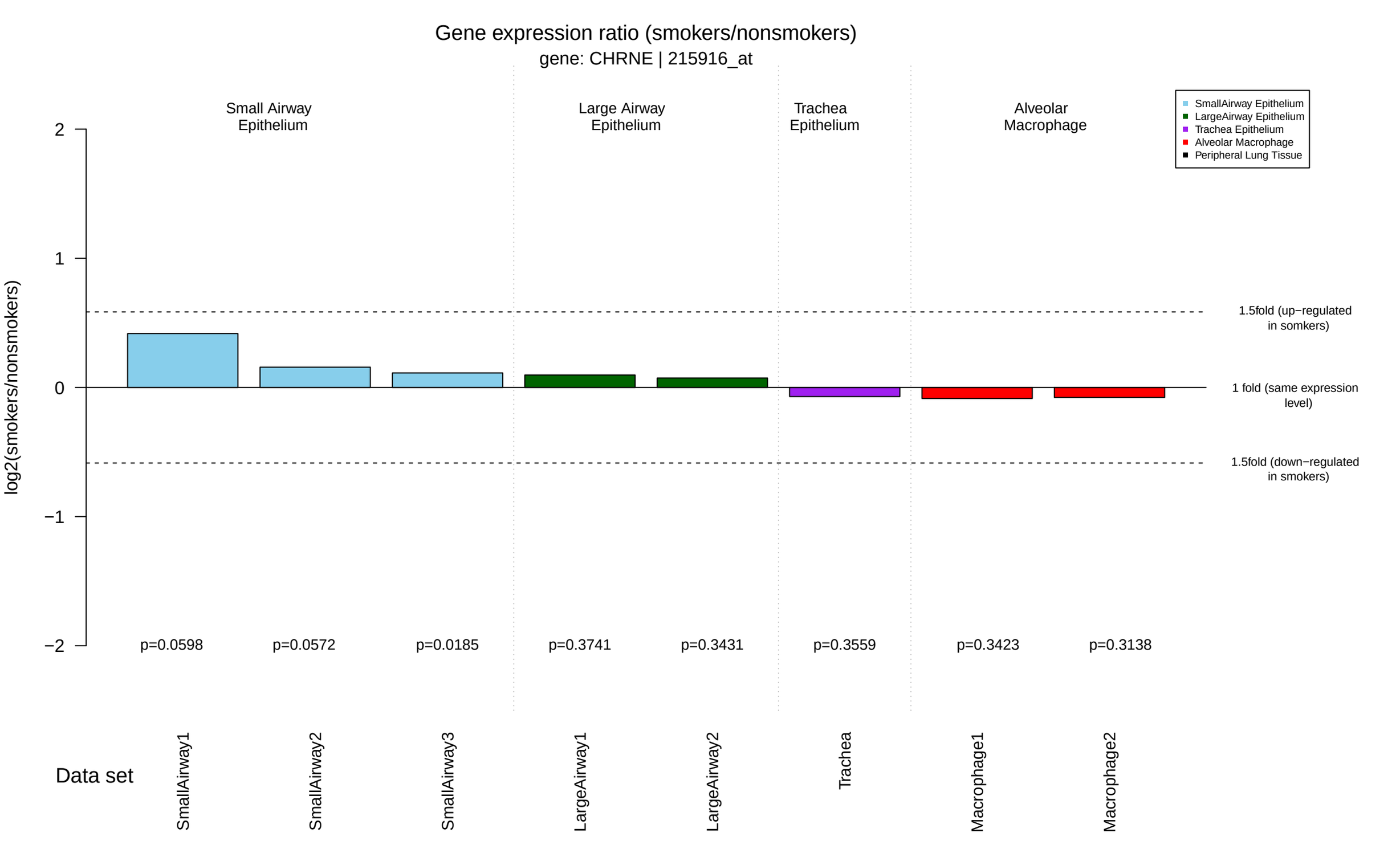
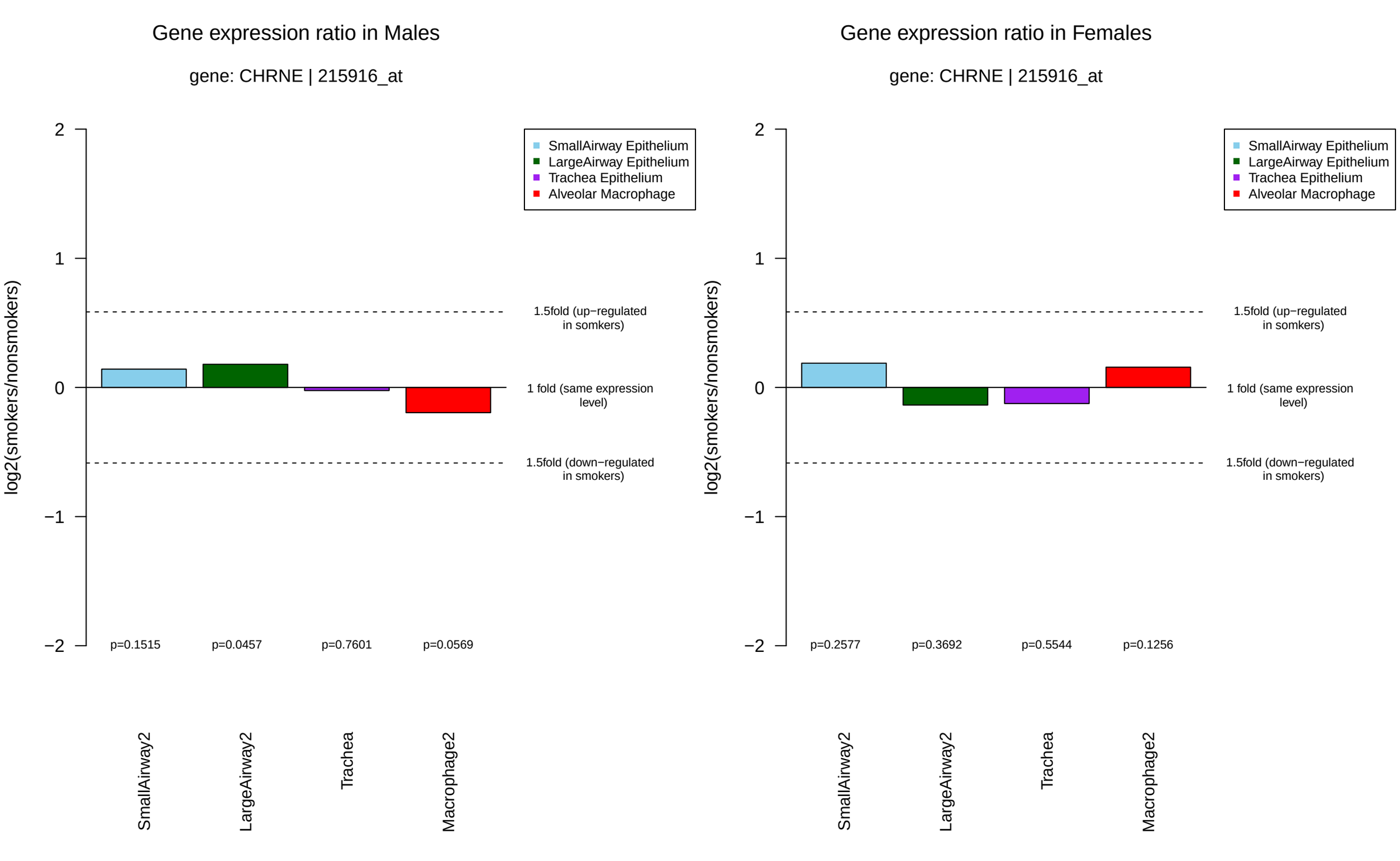
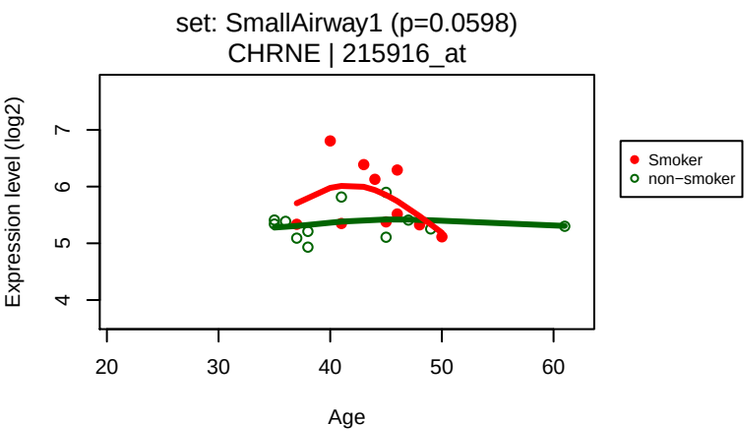
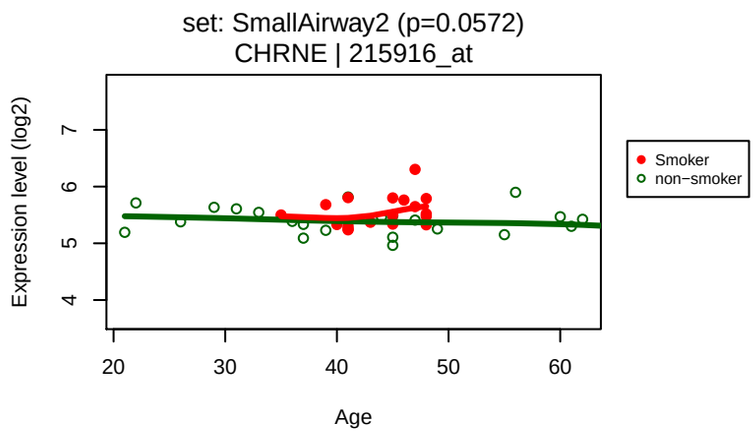
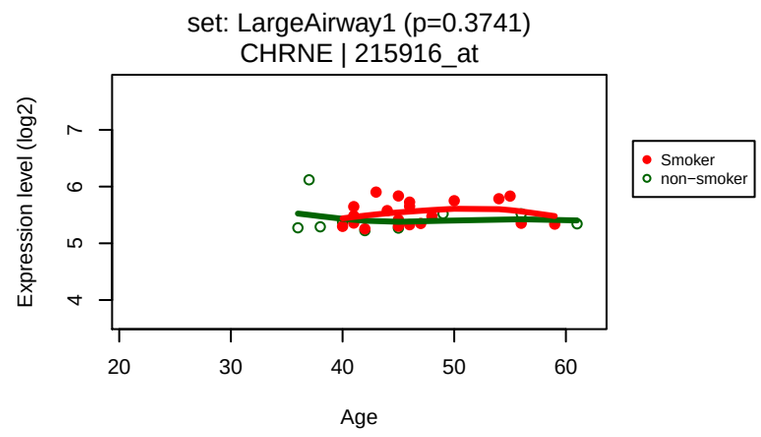

Gene expression of smokers and nonsmokers
gene: CHRNE | 215916_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0598
p=0.0572
p=0.0185
p=0.3741
p=0.3431
p=0.3559
p=0.3423
p=0.3138
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

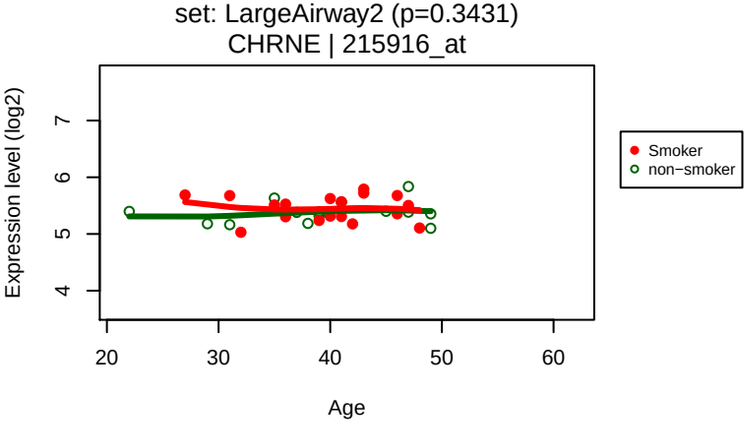

Gene expression ratio (smokers/nonsmokers)
gene: CHRNE | 215916_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0598
p=0.0572
p=0.0185
p=0.3741
p=0.3431
p=0.3559
p=0.3423
p=0.3138
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

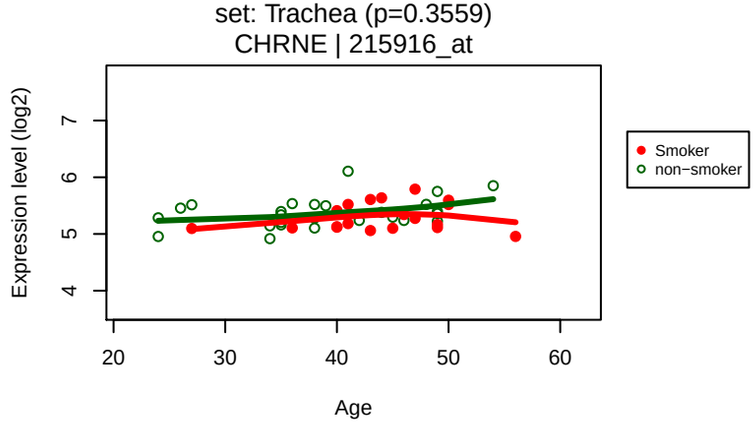

Gene expression ratio in Males
gene: CHRNE | 215916_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1515
p=0.0457
p=0.7601
p=0.0569
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRNE | 215916_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2577
p=0.3692
p=0.5544
p=0.1256
SmallAirway2
LargeAirway2
Trachea
Macrophage2

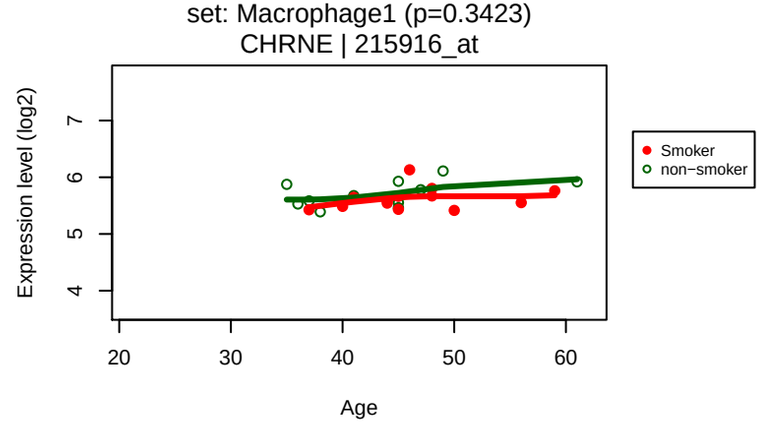

set: SmallAirway1 (p=0.0598)
CHRNE | 215916_at
Expression level (log2)
Age
Smoker
non−smoker

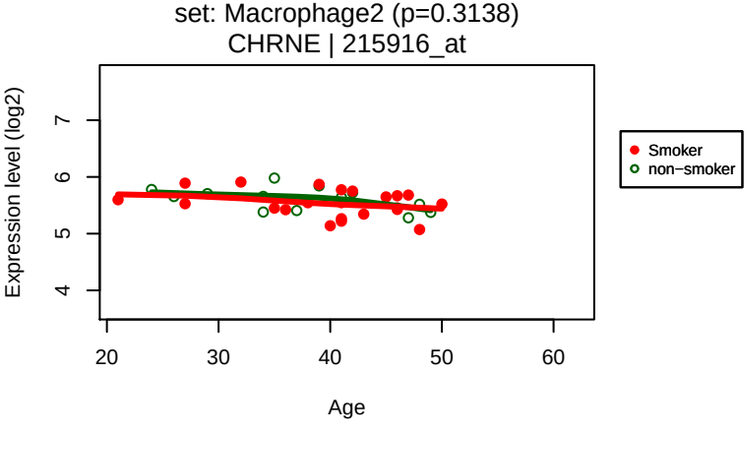

set: SmallAirway2 (p=0.0572)
CHRNE | 215916_at
Expression level (log2)
Age
Smoker
non−smoker

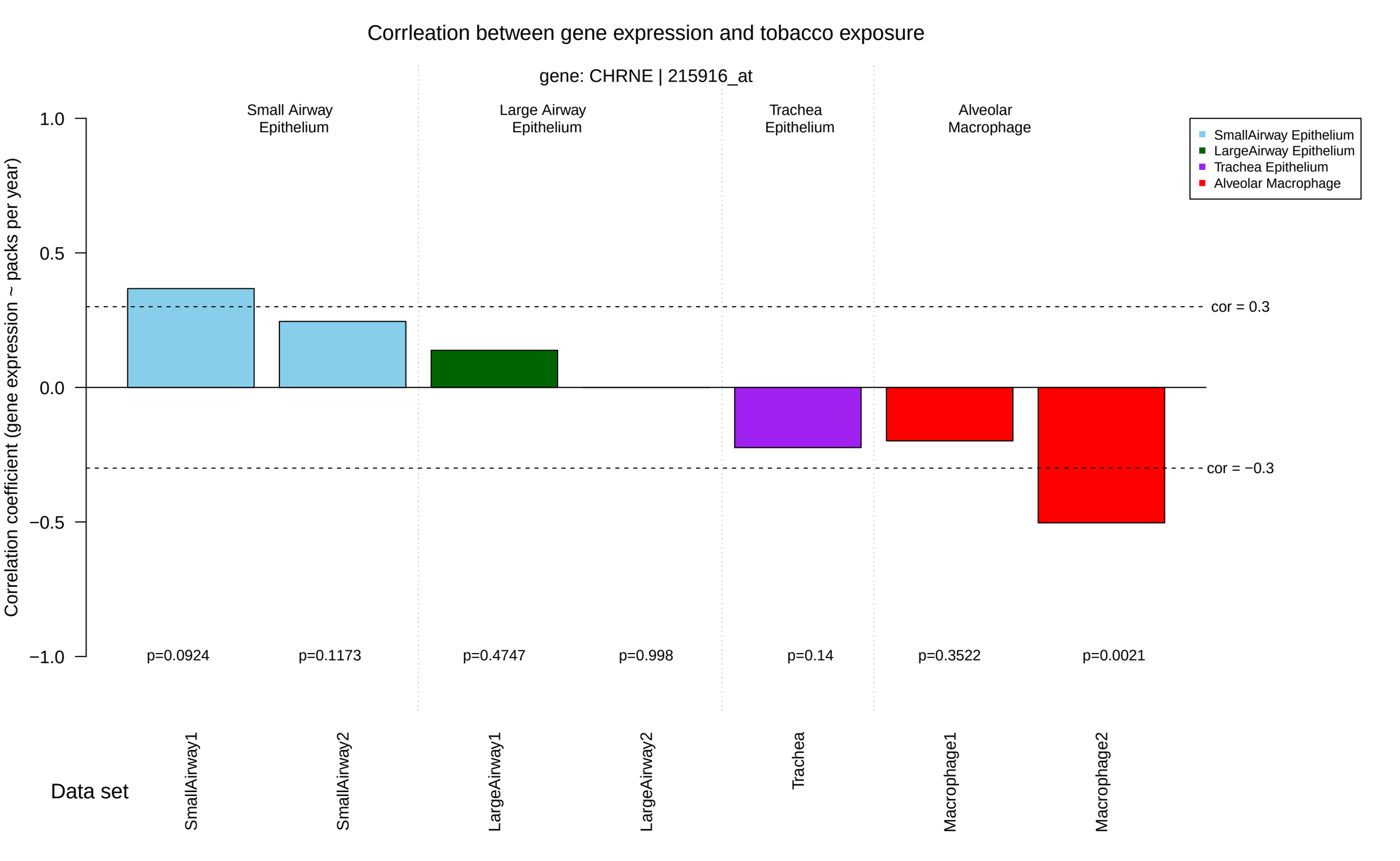

set: LargeAirway1 (p=0.3741)
CHRNE | 215916_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3431)
CHRNE | 215916_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3559)
CHRNE | 215916_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3423)
CHRNE | 215916_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3138)
CHRNE | 215916_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRNE | 215916_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0924
p=0.1173
p=0.4747
p=0.998
p=0.14
p=0.3522
p=0.0021
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2